

MSBA 265 – Foundational Module 1

Interactive EDA, Data Quality Verification, Data Dictionary & Outlier Pipeline

Module 1 Homework Report

Dataset	French Motor Third Party Liability Claims (freMTPL2freq)
Source	OpenML ID 41214
Records analyzed	678,013
Final cleaned records	600,447
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Term	Fall 2026

1. Executive Summary

The French Motor Third Party Liability Claims dataset contains 678,013 policy records and 12 variables. The notebook performs structural checks, boundary audits, business data-dictionary construction, skewness diagnostics, Pearson correlation analysis, distribution visualization, and an outlier audit. The production pipeline applies Tukey's $1.5 \times \text{IQR}$ rule to Population Density and produces a cleaned dataset with 600,447 records.

#	Column	DataType	Count
1	IDpol	float64	non-null
2	ClaimNB	int64	non-null
3	Exposure	float64	non-null
4	Area	str	non-null
5	VehPower	int64	non-null
6	VehAge	int64	non-null
7	DrivAge	int64	non-null
8	BonusMalus	int64	non-null
9	VehBrand	str	non-null
10	VehGas	str	non-null
11	Density	int64	non-null
12	Region	str	non-null

Dataset Overview: 678,013 rows $\times$ 12 columns, no missing values across any feature.

Code script:

```
df.info()
```

2. Part 2 – Raw Boundary Audits & Business Dictionary

2.1 Structural Data and Boundary Verification

Feature	Minimum	Maximum	Audit conclusion
DrivAge	18	100	No negative-age corruption detected
Exposure	0.002732	2.01	No negative exposure detected

BonusMalus	50	230	No negative index values detected
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Negative-value checks in the notebook returned zero for DrivAge, Exposure, and BonusMalus. These checks support the conclusion that the raw dataset contains no negative-age or negative-exposure corruption in the audited fields.

2.2 . Univariate Skewness Diagnostics_Density/Area Defense

According to the dataset, Density is known as a continuous measure of population concentration, while Area represents geographic categories

We calculate the correlation coefficient between Density and Area. A correlation of **0.91** exceeds the workshop's warning threshold of 0.85, indicating **a very strong linear relationship** between the two variables and the potential presence of redundant information. This is a sign of **multicollinearity**; therefore, we consider **removing one of the variables** to avoid introducing excessive redundancy into the model. For this dataset, we removed the “Area” feature to minimize the spatial redundancy and continue.

Feature	Skewness Coefficient	Diagnosis	Action Required
ClaimNb	5.60	Extremely Right-Skewed	Log / Box-Cox Transform
Density	4.65	Extremely Right-Skewed	Log / Box-Cox Transform
BonusMalus	1.73	Moderately Skewed	Log / Box-Cox Transform
VehPower	1.17	Moderately Skewed	Log / Box-Cox Transform
VehAge	1.15	Moderately Skewed	Log / Box-Cox Transform
DrivAge	0.44	Unskewed / Symmetric	Standard Scaler Sufficient
IDpol	0.24	Unskewed / Symmetric	Standard Scaler Sufficient

Exposure	0.09	Unskewed / Symmetric	Standard Scaler Sufficient
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Why do we choose to keep Density instead of Area?

Based on the table above, it is clear to see that the “Density” feature represents the Extremely Right-Skewed by 4.65 in skewness coefficient. Conducting the next step by Blue Print Feature

$$\log(1 + \text{Density})$$

Comparison

Density	Area
Density ↓ Extremely right-skewed ↓ $\log(1 + \text{Density})$ ↓ Predictor X3	Area ↓ Spatial redundancy ↓ Drop

Keep in mind: High correlation → detect redundancy/multicollinearity → then consider business significance and modeling objectives to decide which variables to retain.

2.3 Business Data Dictionary

The notebook defines business meanings for all 12 fields, including policy ID, claim count, exposure, geographic area, vehicle characteristics, driver age, risk score, population density, and administrative region. The executed dictionary export should also be checked before submission because the current notebook output shows only the Region row in the displayed/exported DataFrame, indicating an indentation issue in the dictionary-building loop.

3. Part 3 – Correlation Visual & ClaimNb Decision

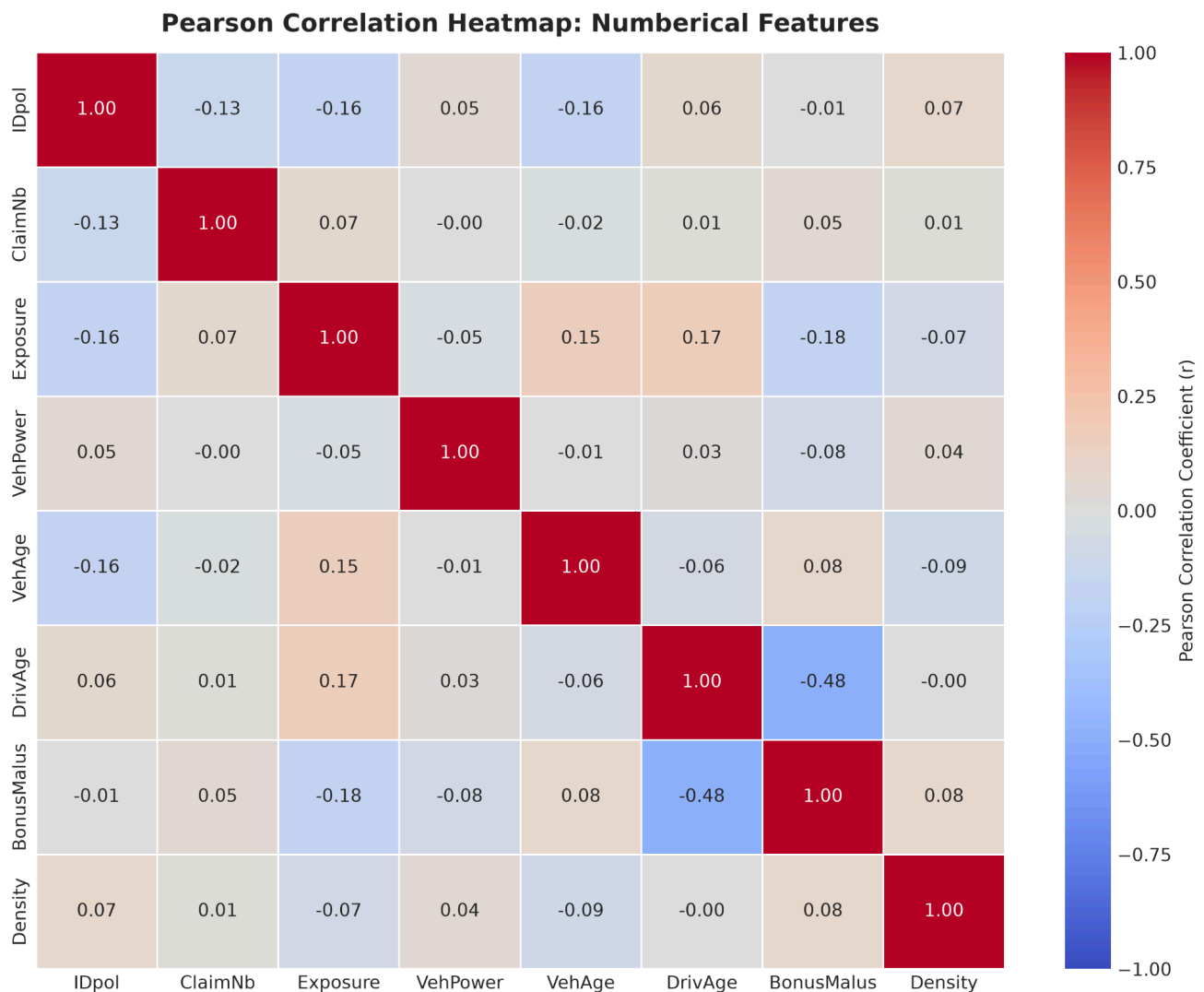


Figure 1. Pearson correlation heatmap generated from the submitted notebook.

What is Pearson correlation purpose?

Pearson $r \in [-1, +1]$

$r \approx +1 \rightarrow$ strong positive linear relationship

$r \approx 0 \rightarrow$ little/no linear relationship

$r \approx -1 \rightarrow$ strong negative linear relationship

Pearson correlation support in providing the prediction which could have risk in serve as a threshold to signal severe multicollinearity.

No severe collinear feature pairs ($|r| \geq 0.85$) were detected across the numerical features. The strongest relationship observed was between DrivAge and BonusMalus ($r = -0.48$), reflecting that older drivers tend to carry lower BonusMalus (risk) scores.

ClaimNb Executive Defense

Question: Why ClaimNb also got the “extreme skewness coefficient” but there are no logarithmic transformations conducted?

Although the automated univariate diagnostics flag ClaimNb with an **extreme skewness coefficient** of **approximately 5.60** and suggest a logarithmic transformation, the rule is intentionally overridden.

More than **95% of policyholders record zero claims**, so a natural-log transformation is mathematically impossible for those observations because $\log(0) = -\infty$. ClaimNb is also a discrete count of insurance events rather than a continuous positive-volume measure.

=> ClaimNb should remain in raw count units and can be modeled as a Poisson rate target with $\log(\text{Exposure})$ used as an offset.

Code script:

- Skewness

```
print ( " ===== " )

print ( " UNIVARIATE SKEWNESS DIAGNOSTICS " )

print ( " ===== " )

numeric_df = df. select_dtypes ( include =[ np. number ])

skewness_series = numeric_df . skew (). sort_values ( ascending = False )

skew_summary = pd. DataFrame ({

    " Feature ": skewness_series .index ,

    " Skewness Coefficient ": skewness_series .values ,

    " Diagnosis ": [ " Extremely Right - Skewed " if s > 3.0 else " Moderately Skewed " if

abs(s) > 1.0 else " Unskewed / Symmetric " for s in skewness_series . values

],

    " Action Required ": [ "Log / Box -Cox Transform " if abs(s) > 1.0 else " Standard Scaler Sufficient " for s in skewness_series . values ])

display ( skew_summary )
```

- Pearson Multicollinearity audit

```
print ( " ===== " )

print ( " PEARSON MULTICOLLINEARITY AUDIT " )
```

```

print (" ===== ")

corr_matrix = numeric_df . corr (). abs ()

upper_tri = corr_matrix . where (np. triu (np. ones ( corr_matrix . shape ),
k=1). astype ( bool ))

high_corr_pairs = []

for col in upper_tri . columns :high_corr_features = upper_tri . index [
upper_tri [col] >= 0.85]. tolist ()

for feat in high_corr_features :

    val = upper_tri .loc[feat , col]

    high_corr_pairs . append ({

        " Feature 1": feat ,

        " Feature 2": col ,

        " Pearson Correlation (|r|)": round (val , 4) ,

        " Status ": " ACTION REQUIRED : Severe Multicollinearity "

    })

if high_corr_pairs :

    display (pd. DataFrame ( high_corr_pairs ))

else :

    print ("[+] No severe collinear feature pairs (|r| >= 0.85) detected
across numerical metrics .")

```

- Pearson correlation Heatmap

```

# Cell 6b: Render Annotated Pearson Correlation Heatmap

plt . figure ( figsize =(10 , 8))

# Compute Pearson correlation matrix across numerical variables

corr = numeric_df . corr ()

```

```

# Generate a diverging color map (Cool - Warm : Blue = Negative , Red =
Positive )

sns.heatmap(

    corr,

    annot=True,

    fmt=".2f",

    cmap="coolwarm",

    vmin=-1.0,

    vmax=1.0,

    linewidths=0.5,

    cbar_kws={"label": "Pearson Correlation Coefficient (r)"}

)

plt . title("Pearson Correlation Heatmap: Numerical Features", fontsize=14,
fontweight='bold', pad=12)

plt . tight_layout ()

heatmap_path = os . path . join ("..", "reports", "figures",
"correlation_heatmap.png")

plt.savefig(heatmap_path, dpi=300)

plt.show()

print(f"[+] Saved correlation heatmap graphics artifact to: {heatmap_path}")

```

4. Part 4 – Distribution Audits & Written Plot Statements

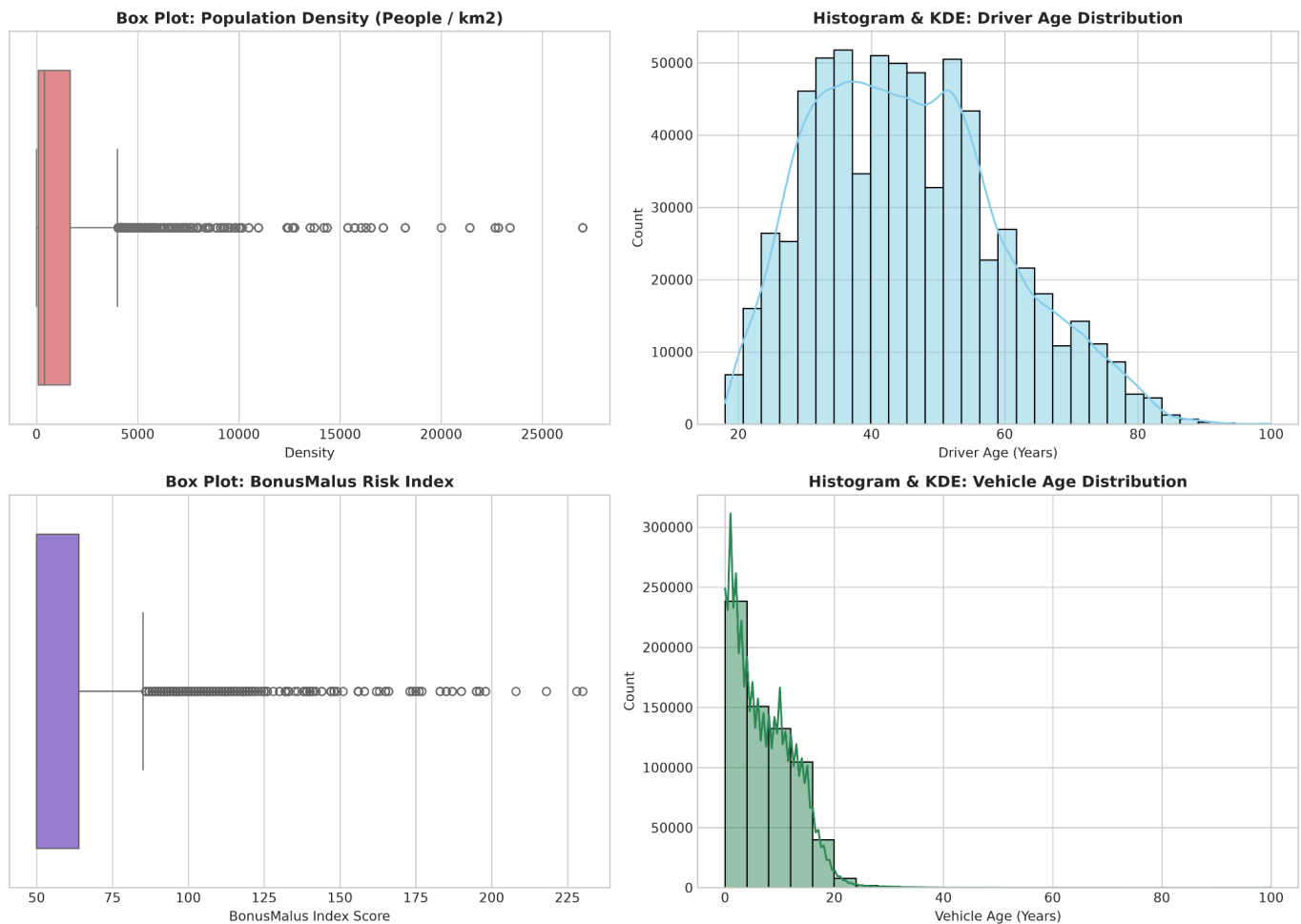


Figure 2. Side-by-side box plot and histogram/KDE distribution audit.

Density – Box Plot

- **Distribution Shape:** Extremely right-skewed; the notebook's skewness diagnostic is approximately 4.65.
- **Key Metrics / Whiskers:** The upper Tukey fence is approximately 4,007 people/km²; observations above the fence include high-density metropolitan locations and are not automatically invalid.
- **Recommended Analytics Action:** Use $\log(1 + \text{Density})$ before linear regression to reduce the influence of the long right tail; use the Tukey rule for the production cleaning step when the assignment requires an outlier-filtered dataset.

DrivAge – Histogram & KDE

- **Distribution Shape:** Approximately symmetric and bell-shaped, with skewness about 0.44.
- **Key Metrics / Whiskers:** Median driver age is 44, with $Q1 = 34$ and $Q3 = 55$.
- **Recommended Analytics Action:** No outlier trimming is required; standard Z-score scaling is sufficient if scaling is needed for a linear model.

BonusMalus – Box Plot

- **Distribution Shape:** Moderately right-skewed, with skewness about 1.73.

- **Key Metrics / Whiskers:** The central distribution is concentrated below 100, with $Q1 = 50$ and $Q3 = 68$; high values represent drivers with elevated risk scores.
- **Recommended Analytics Action:** Keep the variable continuous and scaled; additionally create an Is Surcharge indicator for $\text{BonusMalus} > 100$ when a business interpretation is useful.

VehAge – Histogram & KDE

- **Distribution Shape:** Moderately right-skewed, with skewness about 1.15.
- **Key Metrics / Whiskers:** The distribution peaks among newer vehicles; $Q1 = 2$ years and $Q3 = 12$ years.
- **Recommended Analytics Action:** Bin vehicle age into 0–5, 6–10, and 11+ year groups, or apply standard min-max scaling.

Code Script:

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

sns.boxplot(x=df['Density'], ax=axes[0,0], color='lightcoral')

axes[0,0].set_title('Box Plot: Population Density (People / km2)',
    fontsize=12, fontweight='bold')

axes[0,0].set_xlabel('Density')

sns.histplot(df['DrivAge'], kde=True, ax=axes[0,1], color='skyblue',
    bins=30)

axes[0,1].set_title('Histogram & KDE: Driver Age Distribution',
    fontsize=12, fontweight='bold')

axes[0,1].set_xlabel('Driver Age (Years)')

sns.boxplot(x=df['BonusMalus'], ax=axes[1,0], color='mediumpurple')

axes[1,0].set_title('Box Plot: BonusMalus Risk Index', fontsize=12,
    fontweight='bold')

axes[1,0].set_xlabel('BonusMalus Index Score')

sns.histplot(df['VehAge'], kde=True, ax=axes[1,1], color='seagreen',
    bins=25)

axes[1,1].set_title('Histogram & KDE: Vehicle Age Distribution',
```

```

    fontsize=12, fontweight='bold')

axes[1,1].set_xlabel('Vehicle Age (Years)')

plt.tight_layout()

fig_dir = os.path.join(".", "reports", "figures")

os.makedirs(fig_dir, exist_ok=True)          # <-- creates the folder(s) if
missing

fig_path = os.path.join(fig_dir, "feature_distributions.png")

plt.savefig(fig_path, dpi=300)

plt.show()

print(f"[+] Saved distribution graphics artifact to: {fig_path}")

```

5. Part 5 – Production Pipeline & Final PDF Export

5.1 Production Outlier Pipeline

The submitted production script `src/clean_outliers.py` reads the raw business data, calculates Q1 and Q3 for Density, computes $IQR = Q3 - Q1$, and applies Tukey's $1.5 \times IQR$ lower and upper bounds. Records outside the valid Density interval are removed, and the resulting data are written to `data/cleaned_business_data.csv`.

Pipeline metric	Result
Target feature	Density
Initial dataset	678,013 records
Tukey IQR lower bound	-2,257.00
Tukey IQR upper bound	4,007.00
Outlier records removed	77,566 (11.44%)
Final cleaned dataset	600,447 records

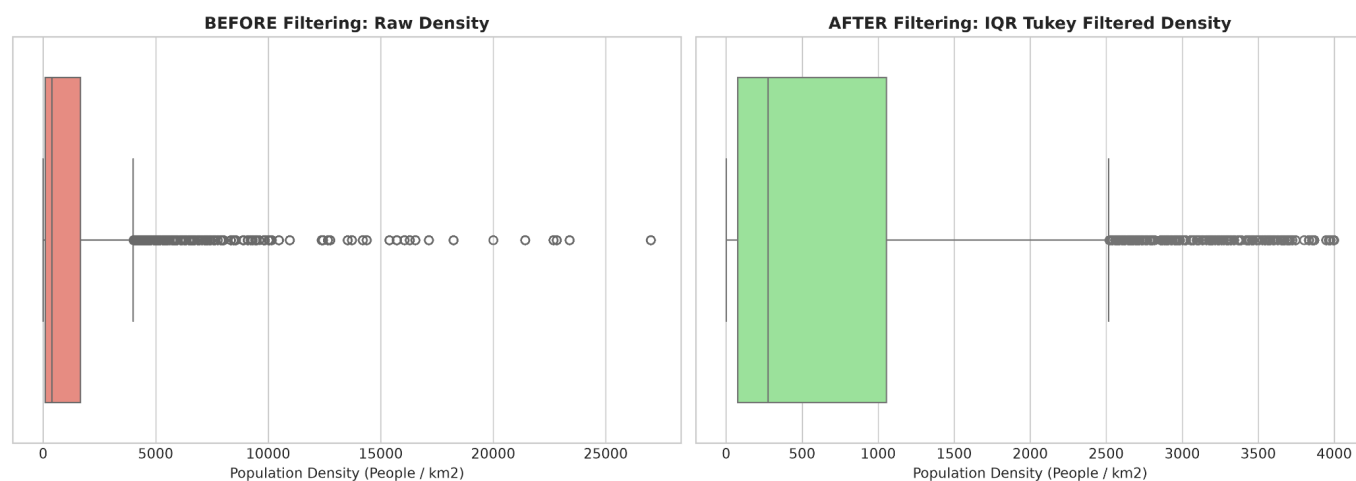


Figure 3. Before/after visual comparison of Density using the Tukey IQR filter.

The production script and the notebook's audit are consistent on the key result: the Tukey filter removes 77,566 observations, or 11.44% of the original dataset, leaving 600,447 records. The resulting CSV is the required cleaned-data artifact for Part 5.1.

Code script:

```
# Cell 8: Mathematical Outlier Filtering & Visual Before/After Comparison

target_feature = "Density"

# Tukey IQR Method

Q1 = df[target_feature].quantile(0.25)

Q3 = df[target_feature].quantile(0.75)

IQR = Q3 - Q1

iqr_lower = Q1 - 1.5 * IQR

iqr_upper = Q3 + 1.5 * IQR

df_iqr_cleaned = df[

    (df[target_feature] >= iqr_lower) &

    (df[target_feature] <= iqr_upper)
```

```

]

# Z-Score Method

z_scores = (

    (df[target_feature] - df[target_feature].mean()) /

    df[target_feature].std()

)

df_z_cleaned = df[abs(z_scores) <= 3.0]

# Outlier Audit Report

print(f"=== OUTLIER AUDIT REPORT FOR '{target_feature}' ===")

print(f"Original Record Count: {len(df):,} records")

print(

    f"Tukey IQR Range [{iqr_lower:.2f}, {iqr_upper:.2f}]: "

    f"Excludes {len(df) - len(df_iqr_cleaned):,} records"

)

print(

    f"Z-Score Threshold (|Z| <= 3.0): "

    f"Excludes {len(df) - len(df_z_cleaned):,} records"

)

# Before vs. After Visualization

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.boxplot(

```

```
x=df[target_feature],

ax=axes[0],

color='salmon'

)

axes[0].set_title(

    f'BEFORE Filtering: Raw {target_feature}',

    fontsize=12,

    fontweight='bold'

)

axes[0].set_xlabel('Population Density (People / km2)')

sns.boxplot(

    x=df_iqr_cleaned[target_feature],

    ax=axes[1],

    color='lightgreen'

)

axes[1].set_title(

    f'AFTER Filtering: IQR Tukey Filtered {target_feature}',

    fontsize=12,

    fontweight='bold'

)

axes[1].set_xlabel('Population Density (People / km2)')

plt.tight_layout()

fig_filter_path = os.path.join(
```

```

    "..",

    "reports",

    "figures",

    "outlier_filtering_comparison.png"
)

plt.savefig(fig_filter_path, dpi=300)

plt.show()

print(

    f"[+] Saved visual outlier comparison artifact to: "

    f"{fig_filter_path}"

)

```

5.2 Final PDF and Reproducibility Checklist

- Final report filename: Module1 Homework Report.pdf
- Production script: src/clean_outliers.py
- Cleaned data output: data/cleaned_business_data.csv
- Notebook: notebooks/01_eda_and_data_dictionary.ipynb
- Figures: correlation heatmap, feature distributions, and outlier-filter comparison.